

Article

The Role of Saudi Arabian Higher Education Institutions in Sustainable Development: Participation, Framework Alignment, and Strategic Insights

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Abstract: Higher education institutions (HEIs) primarily serve as centers of knowledge, capacity builders, and policy influencers and choose to voluntarily engage with the United Nations Sustainable Development Goals (UN-SDGs) as they are not mandated to implement them. In this paper, we examine the role of Saudi Arabian higher education institutions (HEIs) in sustainability efforts through their participation in the Times Higher Education Impact Rankings (THE IR) from 2019 to 2024. Also, the strategic alignment of HEIs with sustainability frameworks is assessed, emphasizing their contributions beyond ranking participation. Using the European University Association (EUA) and the UN-KSA Cooperation Framework (2022–2026), this study identifies both achievements and strategic gaps in HEIs' contributions. Additionally, a comparative analysis with the Saudi Arabian Sustainable Development Report (SDR) of 2024 explores HEIs broader contributions. Hierarchical Cluster Analysis (HCA) is applied to classify HEIs based on SDG engagement, identifying leading, partnering, and supporting HEIs for potential collaboration. The findings reveal a significant increase in HEI participation, rising from 4% in 2019 to 49% in 2024. However, persistent gaps exist, particularly in SDGs 1–2 and 12–15 (addressed by fewer than 10% of HEIs) and SDGs 7, 9–11, and 16 (targeted by fewer than 25% of HEIs). This study highlights the need for a structured strategic alignment beyond rankings, positioning HEIs as enablers of sustainability driven education, research, and institutional governance.



Academic Editor: Giulio Mario Cappelletti

Received: 13 January 2025

Revised: 20 February 2025

Accepted: 10 March 2025

Published: 13 March 2025

Citation: Wajid, H.A. The Role of Saudi Arabian Higher Education Institutions in Sustainable Development: Participation, Framework Alignment, and Strategic Insights. *Sustainability* **2025**, *17*, 2530. <https://doi.org/10.3390/su17062530>

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Keywords: THE Impact Rankings; UN-SDGs; hierarchical cluster analysis; EUA framework; UN-KSA cooperation framework; world rankings; institutional social responsibility; sustainable development; Kingdom of Saudi Arabia

1. Introduction

Over the past few decades, the higher education (HE) system in KSA has undergone significant transformation [1]. Initially, the focus was on establishing a few key HEIs to meet the educational needs [2–4] of the nation. However, with the introduction of Saudi Vision 2030 [5,6], there has been a concerted effort to expand and enhance the quality of HE. This initiative aims to transition the country from an oil-based economy to a knowledge-based one [7]. The Ministry of Education (MoE) has played a pivotal role in this transformation by overhauling curricula and raising education standards nationwide [8]. Notable achievements include the following: (i) the establishment of a total of 67 HEIs (29 public and 38 private) as of 2024; (ii) an increased enrolment rate in both tertiary education (with a gross enrolment ratio of nearly 71% and an average annual growth rate of over 6% since 2012) and higher education (by 2022, approximately 2 million out of 3.3 million Saudis aged 18–24 were enrolled, with 95% attending public sector HEIs);

(iii) the introduction of innovative programs designed to bridge the skills gap between graduates and labor market demands (job placement rate found to be higher in the case of public sector HEI graduates) [8,9]. However, according to the Organization for Economic Co-operation and Development (OECD) Indicators of Education Systems (INES) program, KSA remains one of the lowest among the OECD and partner countries for both tertiary and higher education [10].

A crucial aspect of this transformation is the increasing integration of sustainability, particularly into higher education, addressing 17 global development goals adopted in 2015 by the United Nations (famously known as the UN-SDGs), aimed at creating a prosperous future under the slogan “Transforming Our World” [11,12]. These goals provide a global roadmap for addressing challenges such as poverty, inequality, climate change, environmental degradation, peace, and justice. However, HEIs are not explicitly mandated to deliver outcomes related to these UN-SDGs; rather, they play a pivotal role in advancing sustainability efforts through education, research, and institutional policies. HEIs engagement in SDG-focused initiatives extends beyond rankings by integrating them into their curricula and operations, fostering a generation of graduates committed to sustainable development beyond mere knowledge acquisition [13–16].

In [17], the author highlighted that HEIs cannot act traditionally anymore, as rapid globalization highlights an evident relationship between UN-SDGs (particularly SDG-4) and HEIs, even though SDG outcomes are not obligated upon them. It was further emphasized in [17] that HEIs are expected to reflect leadership in integrating sustainability into education and research. Aligning with UN-SDGs and sustainability practices offers numerous benefits for HEIs, as emphasized in [18], that integrating sustainability into HE enhances HEIs’ visibility and reputation across the globe (leading to attracting talented students, researchers, and faculty members), opens a range of funding and industrial collaboration opportunities, and prepares students for global challenges, inculcating sustainability-oriented skills that are highly essential nowadays, directly impacting their employability. Additionally, aligning with UN-SDGs supports long-term institutional viability and social responsibility, as outlined in [19]. Furthermore, it is studied in [20] that HEIs recognize the strategic importance of embedding sustainable development principles voluntarily into operations. The effectiveness and impact of such voluntary efforts related to UN-SDGs by HEIs is significant in raising awareness and capacity building through the community engagements presented in [21].

A structured framework is essential for helping HEIs systematically integrate sustainable practices into operations, teaching, and research [22]. Therefore, Saudi Arabia has prioritized embedding UN-SDGs into its HE landscapes, positioning HEIs as key drivers of sustainable economic growth and human capital development [23–25], in alignment with Saudi Vision 2030. We recommend readers to consult an extensive report to grip a detailed understanding; specifically, the proactive role HEIs are playing in advancing UN SDGs in relation to Saudi Vision 2030 [26]. This study assesses the performance of HEIs in Saudi Arabia in achieving the UN SDGs using two frameworks specifically designed to address the UN SDGs: (i) the EUA framework [27] and (ii) the UN-KSA Cooperation Framework [28].

This study evaluates the participation, framework alignment (with national and international sustainability frameworks), and performance of Saudi Arabian HEIs in THE IR (2019–2024), addressing the following key research questions:

RQ1: How did the participation of KSA HEIs in covering 17 UN-SDGs in THE IR evolve over time?

RQ2: How well do KSA HEIs align with the European University Association (EUA) framework and the UN-KSA Cooperation Framework (2022–2026)?

RQ3: How does the performance of KSA HEIs in the EUA and UN-KSA Cooperation frameworks correlate with the Saudi Arabian Sustainable Development Report (SDR) of 2024?

This study tests the following hypotheses:

H1: Saudi Arabian HEIs have shown an increasing trend in participation across the 17 SDGs from 2019 to 2024 in THE IR, with varying emphasis on specific SDGs.

H2: The alignment of KSA HEIs with the EUA and UN-KSA frameworks reveals strategic gaps, particularly in SDGs 9 and 16 and Strategic Priority Areas 2–4.

H3: The performance of KSA HEIs in the EUA and UN-KSA Cooperation frameworks correlate positively with the national priorities outlined in the 2024 SDR.

This paper is structured as follows: Section 2 presents an overview of sustainability-focused global rankings and Section 3 presents the adopted methodology. Section 4 analyzes participation trends and achievements of KSA HEIs in THE IR from 2019 to 2024, with a focus on the identification of the top-ranked Saudi Arabian HEIs. Section 5 evaluates HEIs' performances against the EUA framework and UN-KSA Cooperation Framework (2022–2026), identifying alignments and gaps. Section 6 compares the insights of two frameworks with the 2024 SDR, highlighting overlaps and gaps. Section 7 presents a Hierarchical Cluster Analysis (HCA) of HEI SDGs performance, identifying lead, partner, and supporting institutions for collaborative initiatives.

2. An Overview of Sustainability-Focused Global Rankings

Over the past two decades, global rankings such as the Times Higher Education (THE) World University Rankings [29], QS World University Rankings [30], and Academic Ranking of World Universities (ARWU) [31] have been established, evolved, and matured [32]. These rankings aim to evaluate HEIs worldwide using distinct methodologies and focus areas [33,34]. Renowned and leading in their field, these rankings provide a comprehensive overview of HEI performance. To avoid repetition, we recommend consulting the original sources for detailed information [29–31]. However, to highlight their significant role, Figure 1 illustrates that, as of 2024, these rankings cover approximately 8%, 6%, and 10% of the global estimated 25,000 HEIs and approximately 60%, 55%, and 48% of the global estimated 193 countries, respectively.

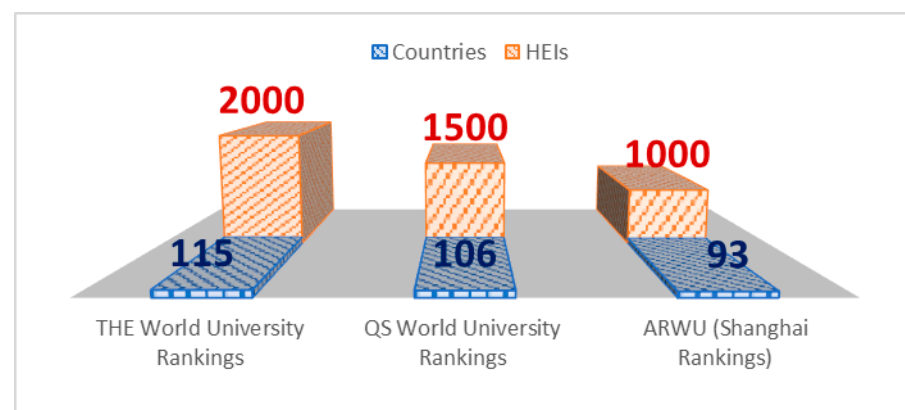


Figure 1. Participation of global HEIs as of 2024–2025 in World University Rankings [29–31].

These rankings offer valuable insights for stakeholders, including educators, policymakers, and students, by highlighting institutional strengths and identifying areas for

improvement. Beyond traditional rankings, there is a growing emphasis on sustainability-focused global ranking systems such as THE IR. These rankings assess institutional performance across 17 UN-SDGs and underscore the increasing importance of social responsibility and sustainable development in HE. Detailed information related to sustainability-focused global assessment platforms is provided in Table 1.

Table 1. Sustainability-focused Global Assessment Platforms for HEIs.

Assessment Platform	Overview	Purpose	Regulatory Authority
[35] STAR, 2010 Sustainability Tracking, Assessment, and Rating System	A transparent, self-reporting framework for HEIs to measure their sustainability performance	To provide a self-reporting tool for assessing sustainability	Association for the Advancement of Sustainability in Higher Education (AASHE)
[36] UIGWUR, 2010 UI GreenMetric World University Ranking	Evaluates HEIs based on sustainability efforts	To rank universities globally by sustainability practices	University of Indonesia
[37] LIFE, 1992 Learning in Future Environments	Supports managing and improving sustainability in HEIs	To promote continuous improvement and social responsibility	Environmental Association for Universities and Colleges (EAUC), UK
[38] HESI, 2012 Higher Education Sustainability Initiative	Enhances HEIs role in advancing sustainable development globally	To promote sustainable development commitments by universities	Partnership between various UN bodies (launched at Rio+20)
[29] THE IR, 2019 Times Higher Education Impact Rankings	Assesses HEIs on teaching, research, stewardship, and outreach aligned with SDGs	To measure university societal impact through SDG metrics	Times Higher Education (THE)
[39] SET4HEL, 2024 Sustainability Evaluation Tool for Higher Education Institutions	An online resource for HEIs to map their contributions to SDGs	To help HEIs measure and improve their SDG impact	UNESCO IESALC in partnership with UNAI and other UN agencies

Table 1 demonstrates that modernizing HEIs requires a shift from merely participating in ranking systems to becoming impactful, mature institutions focused on two core elements [40]: (a) Student Engagement: prioritizing students as primary stakeholders in sustainability initiatives to foster a sense of ownership and responsibility toward sustainable development [41]; (b) Sustainable Campus Models: given their size and population, HEIs resemble small cities, capable of making both positive and negative environmental imprints. To amplify positive impacts, HEIs should adopt responsible environmental practices, such as reducing water, energy, and paper consumption, utilizing renewable energy, promoting recycling, minimizing toxic emissions, implementing sustainable building practices, and increasing green spaces and landscaping [40,41]. Numerous studies further explore campus sustainability strategies within HEIs, highlighting efforts in environmental education, strategic planning, community engagement, and ecological responsibility. These studies often emphasize environmental sustainability over social dimensions [42–48]. Now, in the next section, we briefly introduce THE IR.

Times Higher Education Impact Rankings (THE IR)

THE IR globally assesses HEIs based on their contributions to the 17 UN-SDGs, evaluating universities in four key areas: teaching, research, stewardship, and outreach.

They provide a comprehensive view of how universities are tackling global challenges and striving to improve the world, beyond traditional academic criteria. Since 2019, there

has been a steady increase in the number of HEIs and countries participating in THE IR worldwide. The number of universities has grown from 450 in 2019 to 2152 in 2024, while the number of participating countries has risen from 76 to 125 (see Figure 2). This growth reflects the global commitment of HEIs to align with the UN-SDGs.

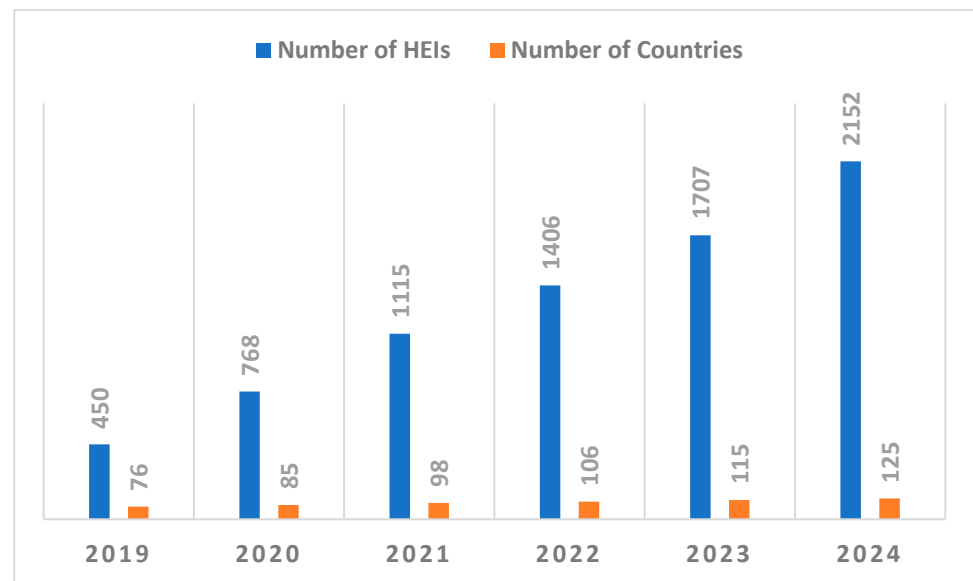


Figure 2. Participation of global HEIs and countries in THE IR since inception.

3. Methodology

3.1. Data Acquisition and Preprocessing

3.1.1. Data Source

The data are inherited from THE IR website link (<https://www.timeshighereducation.com/world-university-rankings>, accessed on 1 December 2024) [29] for the years spanning 2019 until 2024, related to the participating Saudi Arabian HEIs. Yearly data are given in Tables A1–A6 in Appendix A.

3.1.2. Data Preparation

- Missing data, Duplicates, and Outliers: None of the data values were found to be missing, redundant, or extreme.
- Inconsistencies: Each participating HEI in THE IR is evaluated only for four SDGs with SDG-17 as mandatory. Therefore, we standardized yearly data as per total SDGs covered, and participation is indicated by a tick mark. Rows and columns in Tables A1–A6 represent the HEI name and targeted SDG, respectively, to avoid possible inconsistencies.

3.1.3. Data Quantification

Yearly coverage of each SDG is quantified at the bottom of each column as a percentage coverage of participating HEIs to enable statistical analysis for objective measurements and comparison.

3.2. Descriptive Statistical Analysis

It is used to measure the participation of Saudi Arabian HEIs and coverage of UN-SDGs in THE IR.

4. Participation and Performance of Saudi Arabian HEIs in Relation to UN-SDGs in THE Impact Rankings

This section examines the performance of Saudi Arabian HEIs (67 in total: 29 public and 38 private) in relation to the UN-SDGs as assessed by THE IR.

It is clear from Figure 3a that Saudi HEIs first entered these rankings in 2019 with three institutions (two public and one private) which increased to thirty-two institutions (twenty-five public and seven private) in 2024, reflecting substantial growth. Moreover, a linear regression forecast suggests that if this trend continues, all 29 public sector HEIs will be involved in UN-SDG efforts by 2026, whereas 38 private sector HEIs will be on board by 2045. Additionally, it is evident from Figure 3b that the number of SDGs covered by participating HEIs has expanded, from seven SDGs in 2019 to sixteen in 2024, with only SDG-13 remaining unaddressed. Public sector HEIs show strong engagement, with 25 out of 29 participating, compared to just 8 out of 38 private sector HEIs.

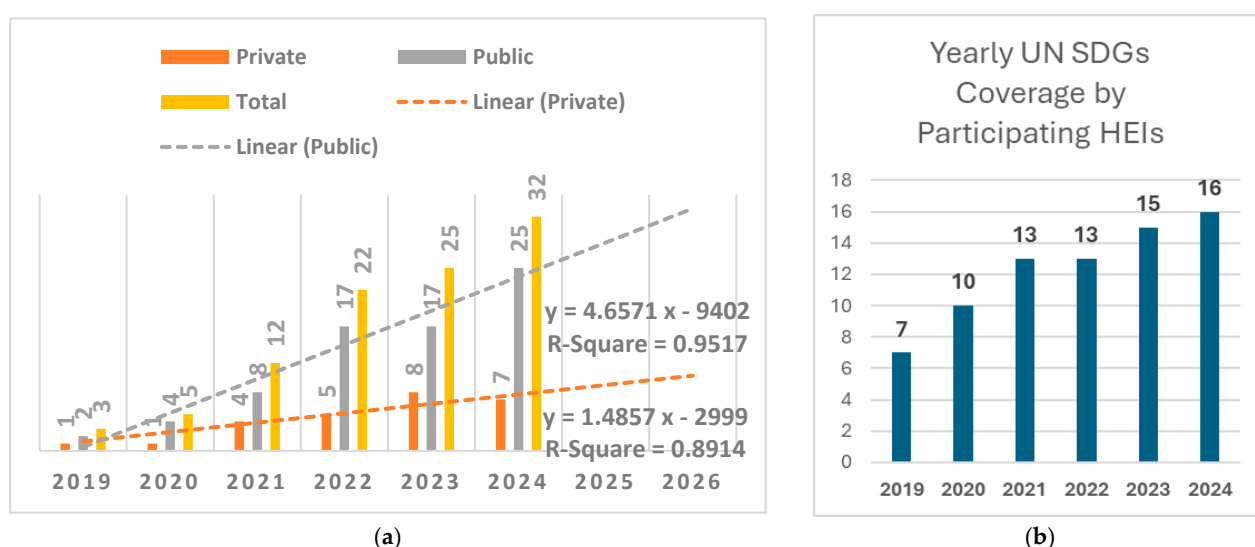


Figure 3. (a) KSA HEIs' participation in THE IR and (b) UN SDGs' coverage since 2019.

In Table 2, a complete list of all public sector HEIs, organized in chronological order by their established year, participation frequency, and the highest achieved ranking bands, i.e., B1 (1–100), B2 (101–200), B3 (201–600), and B4 (600+) are presented. These bands are selected for a compact representation of the data; however, THE IR provides a more detailed distribution of these bands. Additionally, Appendix A contains the comprehensive year-by-year details of achieved bands, targeted UN-SDGs, participating HEIs, and the percentage coverage of the UN-SDGs (see Appendix A Tables A1–A6). Notably, four public sector HEIs including Taibah University, Taif University, Northern Border University, and Shaqra University have not participated in THE IR at all.

Table 3 presents a list of private HEIs that participated in THE IR. Notably, no private institution has reached band B1 (1–100). AFU attained band B2 (101–200) in 2019, followed by PMU, which achieved the same band in 2023–2024. Furthermore, apart from YU, all other institutions have participated at least twice, with only AFU participating all six years, indicating potential for improvement and better rankings in the future (see Table 3). Additionally, participation from 30 private sector HEIs in THE IR is still pending. These institutions are not included here, but we encourage readers to consult the MoE, KSA [8] for more information about them, if needed.

Table 2. List of public sector HEIs of KSA.

Sr. #	Public Sector HEI Name	Short Name	Frequency of Participation in THE IR	Established Year	Highest Achieved Ranking Band/Year
1	Umm Al-Qura University	UQU	4	1950	B3/2021
2	Imam Muhammad bin Saud Islamic University	IMSIU	3	1953	B3/2022–2023
3	King Saud University	KSU	2	1957	B2/2019
4	Islamic University of Medina	IUM	2	1961	B4/2023–2024
5	King Fahd University for Petroleum and Minerals	KFUPM	4	1963	B1/2024
6	King Abdulaziz University	KAU	5	1967	B1/2019–2020, 2022
7	Princess Nora bint Abdul Rahman University	PNU	3	1970	B1/20212
8	Imam Abdulrahman Bin Faisal University	IABFU	4	1975	B2/2024
9	King Faisal University	KFU	4	1975	B1/2024
10	King Khalid University	KKU	5	1998	B3/2022
11	Taibah University	TaibahU	0	2003	Not participated yet
12	Taif University	TU	0	2004	Not participated yet
13	Qassim University	QU	5	2004	B3/2024
14	King Saud bin Abdulaziz University for Health Sciences	KSAU-HS	1	2005	B4/2024
15	Al Jouf University	JU	2	2005	B3/2022
16	Jazan University	JazanU	1	2005	B4/2024
17	University of Hail	UOH	3	2006	B4/2024
18	Al Baha University	BU	3	2006	B4/2024
19	Najran University	NU	2	2006	B4/2023
20	University of Tabuk	UT	3	2006	B4/2024
21	Northern Borders University	NBU	0	2007	Not participated yet
22	Prince Sattam Bin Abdulaziz University	PSAU	3	2008	B4/2022
23	King Abdullah University of Science and Technology	KAUST	3	2009	B2/2022–2023
24	Majmaah University	MU	1	2010	B4/2024
25	Shaqra University	SU	0	2010	Not participated yet
26	Saudi Electronic University	SEU	3	2011	B3/2022
27	University of Bisha	UB	2	2014	B4/2023
28	University of Hafr Albatin	UOHB	1	2014	B4/2024
29	University of Jeddah	UJ	3	2014	B4/2023

Table 3. List of private sector HEIs of KSA.

Sr. #	Private Sector HEI Name	Short Name	Participation Frequency	Founding Year	Highest Achieved Ranking Band/Year
1	Prince Sultan University	PSU	4	1998	B3/2022
2	Effat University	EU	2	1999	B4/2024
3	University of Business and Technology	UBT	3	2000	B4/2022
4	Al Yamamah University	YU	2	2001	B4/2023
5	Alfaisal University	AFU	6	2002	B2/2019
6	Batterjee Medical College	BMC	2	2005	B4/2023
7	Prince Mohammad Bin Fahd University	PMU	3	2006	B2/2023–2024
8	AlMaarefa University	UM	4	2009	B3/2022–2023

Also, it is evident from Tables 2 and 3, a maximum of 36% of HEIs were classified in band B4 until 2022. However, starting in 2023, this trend reversed, with over 60% of HEIs qualifying in band B4 due to increased participation. Additionally, since 2020, the majority of HEIs have been placed in the lower bands, B3 and B4. This indicates that only a small number of participating HEIs rank among the top 200 globally.

Also, it is clear from Figure 4 that over 50% of established HEIs have not yet participated in THE IR, including four public and thirty private. Additionally, AFU is the only HEI that participated in every edition of THE IR. Moreover, only 6–16% of the HEIs have participated one to three times, while 9% and 4% have participated four and five times, respectively, which is critically low.

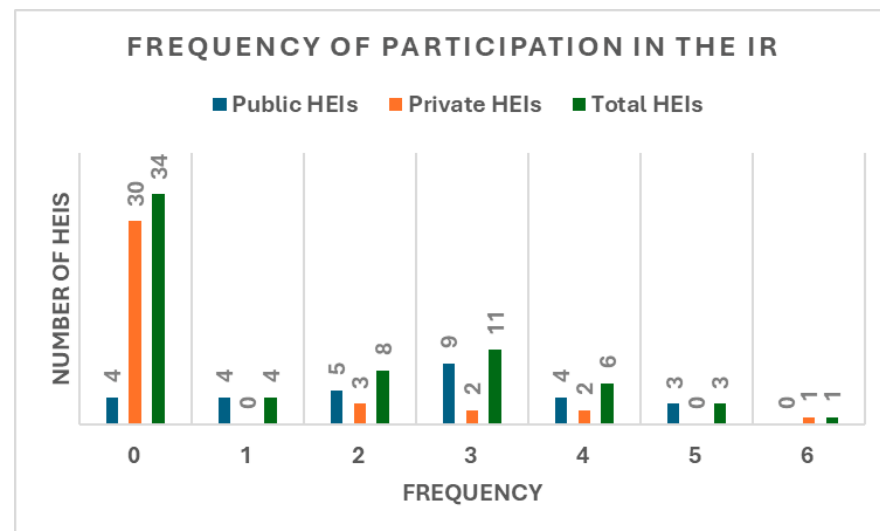


Figure 4. Participation frequency of HEIs in THE IR.

4.1. Top 100 Ranked HEIs of Saudi Arabia in THE IR

In this section, we evaluate the progress of the top 100 ranked HEIs, specifically KAU, PNU, KFUPM, and KFU, in THE IR, with details given in Table 4. These institutions represent only 6% of the total 67 HEIs in Saudi Arabia and nearly 13% of those that participated in THE IR. Notably, KAU and PNU were among the founding institutions and have consistently achieved impressive global rankings within the top 100.

Table 4. Top 100 ranked Saudi Arabian HEIs in THE IR from 2019 to 2024.

Year	HEIs Name and Achieved Rank
2019	KAU (54th)
2021	KAU (46th), PNU (27th)
2022	KAU (4th)
2024	KFUPM (81st), KFU (99th)

KAU ranked 54th in the inaugural year of 2019 and maintained its position with the 46th ranking in 2021 and an impressive 4th position in 2022. Meanwhile, PNU reached the 27th position in 2021. In 2024, KFUPM and KFU achieved the rankings of 81st and 99th, respectively. Each of these universities was established in the 20th century and has the potential to lead and inspire the remaining HEIs in KSA.

4.2. UN-SDGs Coverage by B1-Ranked HEIs of KSA

Now, we discuss the coverage of UN-SDGs by four B1-ranked HEIs over the years, as detailed in Table 5.

Table 5. UN-SDGs covered by top 100 ranked HEIs of KSA.

HEIs Name	UN-SDGs Covered																
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
KAU				✓	✓			✓	✓	✓	✓					✓	✓
PNU				✓	✓			✓		✓							✓
KFUPM						✓	✓	✓	✓					✓			✓
KFU	✓	✓		✓		✓	✓	✓			✓	✓					✓
Percentage coverage by participating HEIs	25	25	0	75	50	50	50	100	50	50	50	25	0	25	0	25	100

Since SDG-17 is mandatory, all four institutions have addressed it, along with only SDG-8. Additionally, three of the institutions have focused on SDG-4. Furthermore, KAU and PNU both covered SDGs 4, 5, 8, and 10, while KFUPM and KFU both targeted SDGs 6, 7, and 8. SDG-9 is covered by KAU and KFUPM, and SDG-11 by KAU and KFU. Lastly, SDGs 1, 2, 12, 14, and 16 are each covered by only one of the four institutions. However, none of these four institutions addressed SDGs 3 and 13. In conclusion, SDGs 4 to 11 represent the collective strength of these four institutions. Their efforts in these areas can serve as a benchmark for the other HEIs of KSA to follow. None of the private sector HEIs of KSA have achieved B1 (1–100) band rankings so far.

4.3. Key Findings and Conclusion for Hypothesis 1

In this section, we present the key findings (denoted by KF) and conclude Hypothesis 1.

KF-1: In 2019, only three HEIs were involved, expanding to thirty-two HEIs by 2024, covering sixteen UN SDGs, with only SDG 13 remaining unaddressed. Public sector HEIs lead in this trend, showing stronger engagement (25 out of 29) than private HEIs (8 out of 38). However, it is commendable that most of the private sector HEIs are quite young and are instead committed to contributing to UN-SDGs related efforts with AFU, consistently participating in all six editions of THE IR. Also, participation by the remaining 34 HEIs (4 public, 30 private) will amplify the overall progress related to UN SDGs.

KF-2: Four public-sector HEIs (KAU, AFU, KFUPM, and PNU) achieved top 100 rankings in THE IR six times. Also, KAU achieved the 4th rank in the year 2022. These HEIs have demonstrated leadership qualities across UN-SDGs 4–11. However, B1-ranked HEIs are only 6%, and all of these are public sector HEIs. Following, four out of eight private sector HEIs (namely, AFU, PSU, PMU, and BMC) are identified as being very consistent over the years and are potential candidates to qualify for B1 rankings. Similarly, two public sector HEIs, namely UQU and PSAU, must also plan for B1 rankings.

KF-3: Certain HEIs are positioned to lead in specific SDGs based on consistent performance and qualified higher rankings in THE IR, such as KFUPM and KFU in SDGs 6–8, and KAU and PNU in SDGs 4–5, 8, and 10.

Concluding, the participation of Saudi Arabian HEIs in addressing the UN SDGs has been remarkable based on a significant upward trend in participation and achieving higher rankings across a broad range of UN-SDGs, supporting Hypothesis 1 of this study.

5. Assessment of Framework Alignment in Saudi Arabian HEIs

In this section, we evaluate the significant achievements and strategic gaps in the alignment of Saudi Arabian HEIs with the UN-SDGs. This assessment is based on two frameworks: (a) an international framework (the EUA framework) and (b) a national framework (the UN-KSA Cooperation Framework: 2022–2026).

5.1. Alignment with an International Framework: EUA Framework

The European University Association (EUA), established in 2001, represents and supports higher education institutions across 48 countries [27]. The EUA plays a crucial role in shaping European higher education policies. Since 2015, the EUA has recognized the importance of the UN SDGs and has committed to working towards these goals both within Europe and with its global partners, equipping students to address global challenges effectively.

The EUA strongly advocates for HEIs to embed the UN-SDGs into their core strategies. It places particular emphasis on four key SDGs: SDG-4 (providing quality education), SDG-9 (fostering innovation through cutting-edge research), SDG-16 (promoting good governance and becoming pillars of civil society), and SDG-17 (building global and local partnerships). HEIs' contributions to these four SDGs are essential for advancing the achievement of the remaining goals [27].

Figure 5 illustrates the progress of KSA HEIs according to the EUA framework, focusing on four key SDGs. Starting with SDG-4, only three HEIs participated in 2019, with just one (33%) targeting this goal. Growth has been slow, and by 2024, only 59.4% of participating HEIs (19 out of 32) focused on SDG-4. Even assuming many of these institutions are likely research oriented, this is not yet reflected in their performance on SDG-9. Alarming, a downward trend is observed for SDG-9, with only 12.5% (four out of thirty-two) of HEIs targeting it by 2024. As for SDG-16, the data suggest that HEIs have not shown strong commitment, with maximum participation reaching only 20% over the years. SDG-17, being mandatory, has been addressed by all participating HEIs.

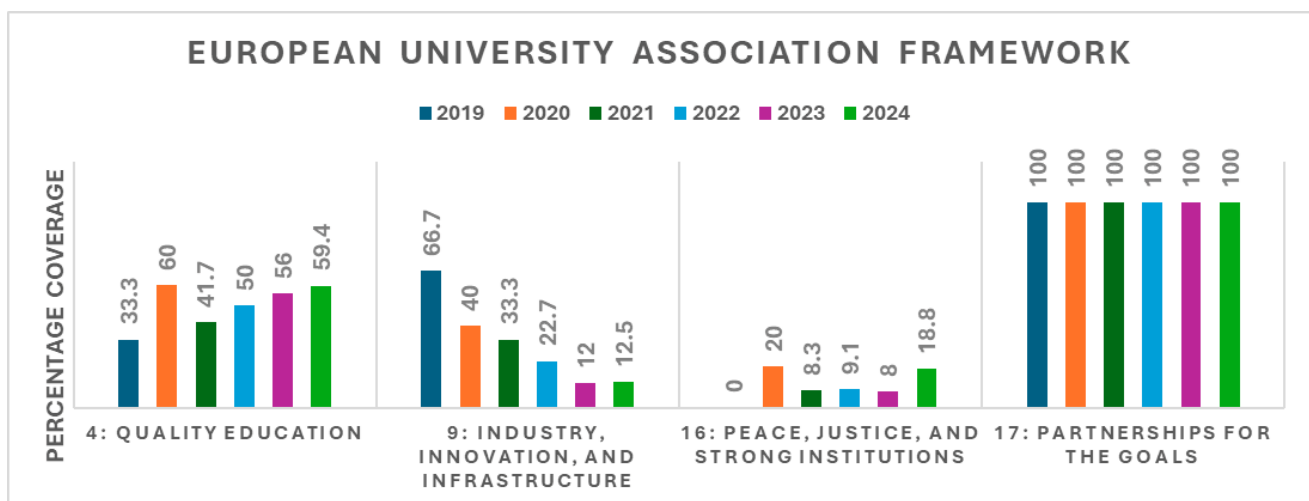


Figure 5. KSA HEIs' performance in the four key UN SDGs identified by EUA framework [27].

Additionally, we observe the participation of leading HEIs in relation to the essential SDGs outlined by the EUA framework. Table 5 shows that only KAU has targeted all these critical SDGs. PNU and KFUPM also addressed SDG-4, while KFUPM is also focused on SDG-9. Figure 5 summarizes participation in 2024, indicating that 59% of all participating HEIs targeted SDG-4, 13% focused on SDG-9, and 19% on SDG-16. There is a strong need for greater commitment from all participating HEIs, including the leading institutions, to effectively target these SDGs for a lasting impact.

5.2. Alignment with a National Framework: UN-KSA Cooperation Framework (2022–2026)

In this section, we evaluate the progress of KSA HEIs from 2019 to 2024, reflecting their priority on specific UN-SDGs through the lens of THE IR in relation to the four Strategic

Priority Areas of the United Nations Sustainable Development Cooperation Framework with the Government of the Kingdom of Saudi Arabia (2022–2026) [28].

5.2.1. Strategic Priority Area 1: People

Strategic Priority Area 1 (SPA-1) emphasizes the importance of enabling all individuals to realize their potential with dignity, equity, and equality in a healthy environment, guided by the slogan “Leaving No One Behind” (LNOB). This priority area targets SDGs 1–5 [28].

Figure 6 clearly illustrates the participation trends of HEIs in relation to the UN SDGs. For SDGs 1 and 2, participation has remained extremely low over the years, with only 6% and 3% of HEIs engaged in these goals by 2024. In contrast, SDG-3 shows a significant level of engagement, with nearly 50% of HEIs participating in 2024, reflecting a strong emphasis on health initiatives. SDG-4 demonstrates substantial progress, with participation rising from approximately 33% in 2019 to about 60% in 2024. Lastly, for SDG-5, HEIs showed notable participation, starting with 66.7% in 2019 and settling to just above 37% by 2024. SDGs 1 and 2 urgently require attention. Notably, among the institutions ranked in the top band, B1 (1–100), only KFU targeted these two goals, and that was only in the year 2024. Surprisingly, none of band B1 institutions targeted SDG-3.

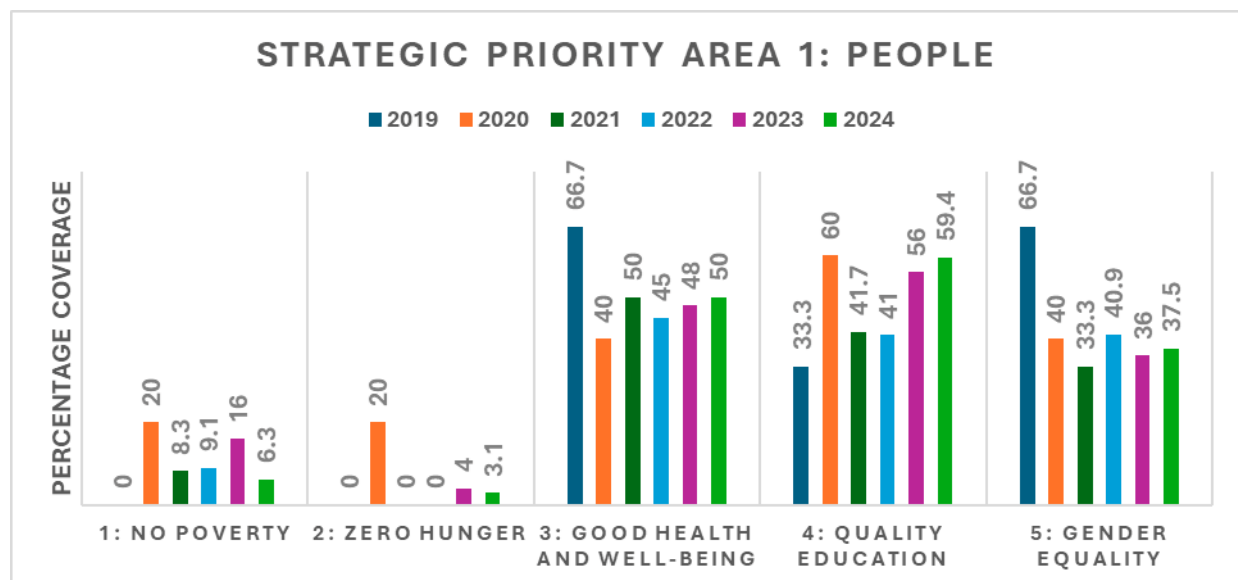


Figure 6. KSA HEIs’ performance related to SPA-1 through the lens of THE IR.

5.2.2. Strategic Priority Area 2: Planet

Strategic Priority Area 2 (SPA-2) aims to protect the planet from degradation by promoting sustainable consumption and production practices. It emphasizes the sustainable management of natural resources and calls for urgent action on climate change, targeting SDGs 7–11 [28].

Figure 7 illustrates the performance of HEIs in relation to SPA-2. In 2021, 25% of HEIs participated in SDG-7, but this participation gradually declined to 15.5% by 2024. This is concerning, as HEIs worldwide are striving to become green or sustainable campuses. Small-to-medium-scale initiatives must be undertaken by KSA HEIs to address this gap. For SDG-8, steady participation is evident, slightly decreasing from 40% in 2020 to nearly 37.5% in 2024. SDG-9 showed a significant decline, with participation dropping from 66.7% in 2019 to 12.5% by 2024. SDG-10 had a below-expectation trend, with a maximum participation of 33% in 2019 declining to 25% in 2024. Unfortunately, SDG-11 was not targeted by any HEIs from 2020 to 2023, with a critically low participation of 6% in 2024. SDGs 7 and 11 require significant attention from all HEIs to fully realize the potential of

SAP-2. Among band B1 institutions in KSA, at least two out of four institutions targeted SDGs 7–11, including SDGs 7 and 11. Specifically, KFUPM and KFU focused on SDG-7, while KAU must reach out to other HEIs to enhance coverage for SDG-11. Furthermore, KAU and PNU can mobilize other HEIs specifically targeting SDG-10.

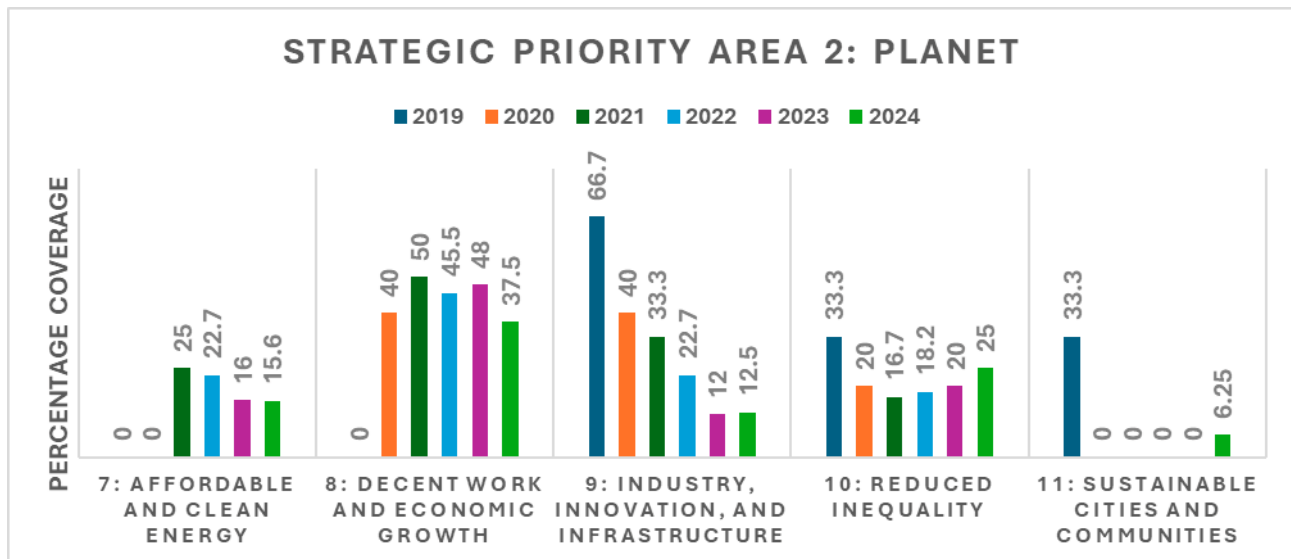


Figure 7. KSA HEIs' performance related to SPA-2 through the lens of THE IR.

5.2.3. Strategic Priority Area 3: Prosperity

SPA-3 aims to ensure that all people can enjoy prosperous and fulfilling lives, focusing on SDGs 6 and 12–15 [28]. Starting with SDG-13, none of the HEIs have covered this goal so far. Similarly, SDG-15 has seen no participation since 2022, with critically low participation, peaking at just 4%, as shown in Figure 8. Interestingly, none of the HEIs targeted all SDGs under SPA-3 in 2019–2020. For SDGs 6, 12, and 14, the highest participation rates were 18.2%, 9.1%, and 12%, respectively. Overall, participation in SPA-3 is very poor, indicating a need for better planning and increased and collaborated efforts. Moreover, even all the top ranked band B1 institutions do not cover SDGs 13 and 15. SDGs 12 and 14 are covered only once by KFU and KFUPM. Finally, SDG-6 is covered by both KFU and KFUPM (see Table 5).

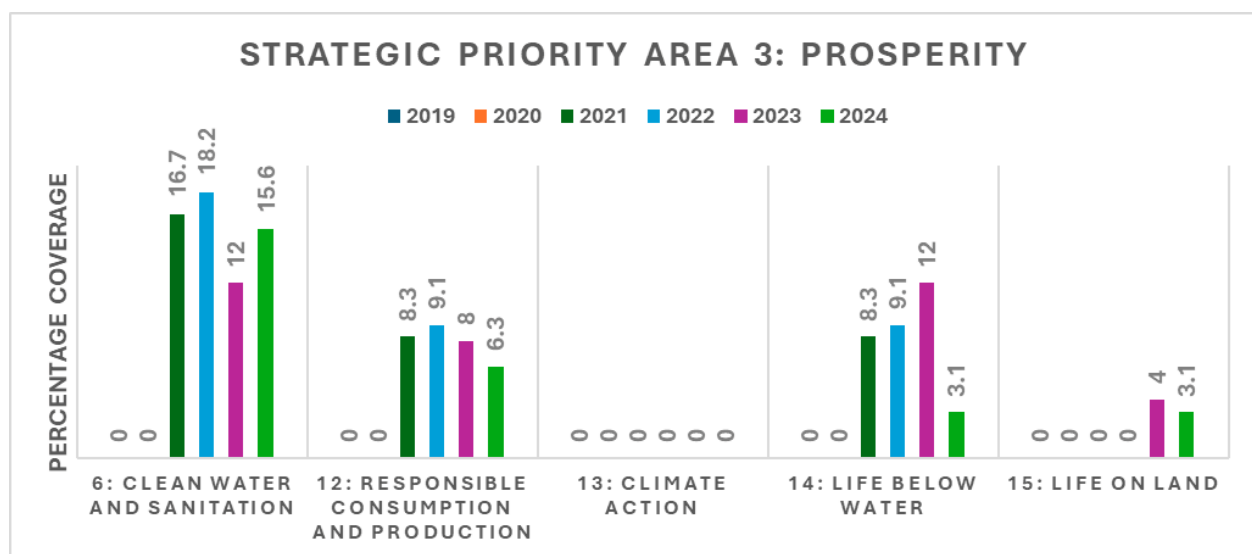


Figure 8. KSA HEIs' performance related to SPA-3 through the lens of THE IR.

5.2.4. Strategic Priority Area 4: Peace, Partnership, and Other Cross-Cutting Issues

SPA-4 aims to promote peaceful, just, and inclusive societies by mobilizing the necessary resources to achieve SDGs 16 and 17 [28]. However, there is a notably low participation rate in SDG-16 over all the years. Even in the year 2024, with thirty-two HEIs on board, only six HEIs targeted this SDG-16, as shown in Figure 9. Also, only KAU from the B1-ranked institutions targeted SDG-16.

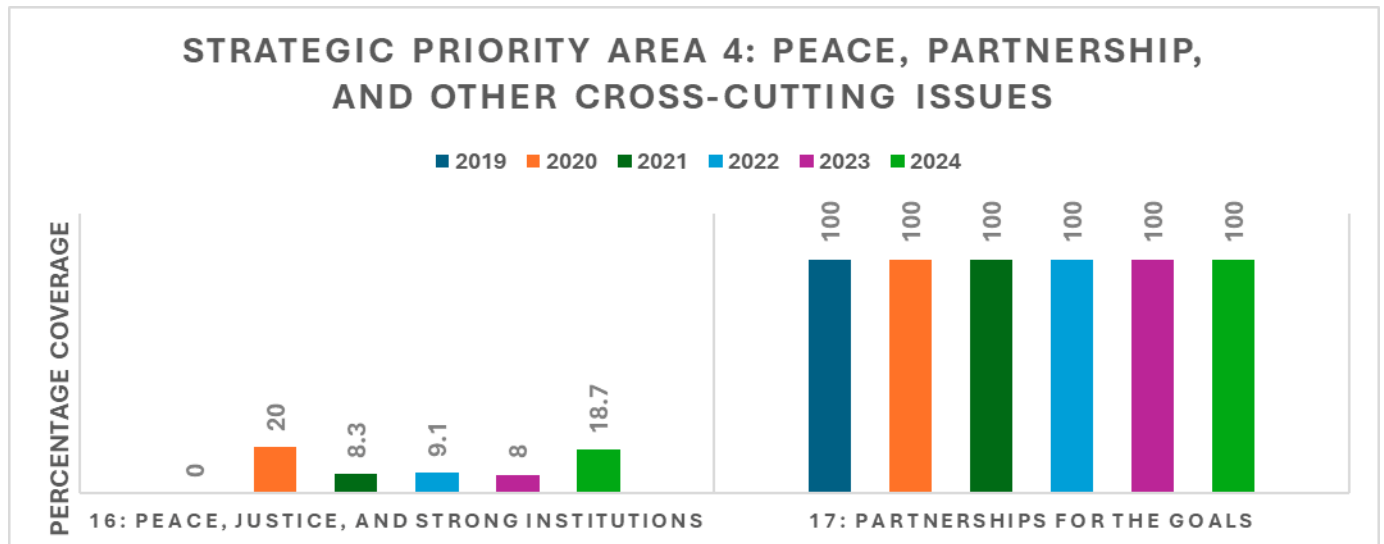


Figure 9. KSA HEIs' performance related to SPA-4 through the lens of THE IR.

5.3. Key Findings and Conclusion for Hypothesis 2

In this section, we now present key findings and conclude Hypothesis 2. The UN-SDGs targeted by HEIs are classified into three categories given below:

(i) Low Priority (percentage coverage of less than 25)

A total of 11 out of 17 (65%) of the UN-SDGs, including 1–2, 6–7, and 10–16, was targeted by less than 25% of the participating HEIs the majority of the time over the years. Moreover, SDG-9 started with a high participation trend; however, it faced a declining trend after 2021, staying below 25%.

(ii) Moderate Priority (percentage coverage of 26–50)

A total of four out of seventeen (23.5%) of the UN-SDGs, including 3, 5, 8–9 was targeted in this range.

(iii) High Priority (percentage coverage of more than 50)

Only SDG-4 qualified in this category.

Concluding, none of the HEIs have prioritized SDG-13, reflecting a gap in addressing climate-related goals. Also, there is limited HEI engagement related to SDG-1 (No Poverty) with coverage only by KFU amongst higher ranking institutions. Moreover, SDGs 9 and 16 and Strategic Priority Areas 2–4 require considerable attention and demand a responsible, committed, and collaborative role by HEIs for strong alignment to ensure smoother attainment of the key pillars of Saudi Arabian Vision 2030. Based on the above key findings, Hypothesis 2 is fully supported because of the evident strategic gaps and minimal achievements.

6. Comparative Analysis of UN-SDG Attainment: Insights from the EUA and UN-KSA Frameworks and the Saudi Arabian SD Report (2024)

The SDG Index and Dashboards annually assess SDG progress in 193 UN member states using 125 indicators, including 98 global and 27 specific to OECD countries. Only countries with less than 20% of missing data are scored, covering 167 nations this year (2024), including Guinea-Bissau and new BRICS plus members, which include KSA. For detailed data and methodology, visit the SDG Transformation Center website [49,50].

Figure 10 presents the progress and status of all 17 UN-SDGs for the year 2024 with distinctive colors, arrow mark directions, and dots.



Figure 10. Saudi Arabian UN-SDG attainment progress in SDR-2024 (courtesy of [50]).

6.1. Comparison of EUA Framework Insights with the Saudi Arabian SD Report (2024)

It is clear from Figure 10 that all key SDGs (4, 9, 16, and 17) of the EUA framework are still facing moderate to significant challenges; however, only SDGs 9 and 17 are on track, due to the key initiatives launched by governmental institutions, particularly the proactive role of MCIT. For more details, readers are encouraged to consult Saudi Arabia's comprehensive 2024 annual report titled "Digital & Space Sustainability" [51]. There is a notable upward trend in internet usage, mobile broadband subscriptions (per 100 population), and an above-average score of 3.6 in the Logistics Performance Index. SDG-4 is moderately improving, and SDG-16 shows a declining progress trend. Moreover, interestingly, the top three universities achieved an average score of 53/100 in THE IR in 2024 [50]. These countrywide findings are in good agreement with the insights given in Section 5.1. As the number of HEIs increases, participation in SDG-4 also rises, with 60% of participating HEIs targeting this goal in 2024. However, 72% of all HEIs (48 in total: 13 participatory and 35 non-participatory) have not targeted SDG-4, highlighting a significant challenge. Re-quoting from Section 5.1, in 2024, only 59%, 13%, and 19% of participating HEIs targeted SDGs 4, 9, and 16, respectively, revealing a positive correlation with KSA SDR 2024.

6.2. Comparison of UN-KSA Framework Insights with the Saudi Arabian SD Report (2024)

6.2.1. SPA-1

We also examine countrywide progress in relation to SPA-1, though data for SDG-1 are missing. Major challenges for SDGs 2 and 5 have resulted in stagnant progress, while SDGs 3 and 4 show moderate improvement, though substantial challenges remain (see Figure 10). Notably, progress by HEIs aligns closely with national progress on SDGs 1–5, highlighting a clear opportunity for HEIs to play a strategic role in closing these gaps.

6.2.2. SPA-2

Data for SDG-10 are missing in the countrywide progress report. Major challenges exist for SDGs 7 and 11, with SDG-7 showing moderate improvement and SDG-11 remaining stagnant. SDG-8 is on track but still faces significant challenges (see Figure 10). Notably, the progress made by HEIs closely aligns with national progress for SDGs 7–11. It is crucial for both HEIs and national governing bodies to target these SDGs.

6.2.3. SPA-3

There is a positive correlation between the nationwide performance of SDGs 12–15 and the priority given to them by HEIs. Major challenges exist for SDGs 12–14, with SDGs 12 and 13 showing moderate improvement, while SDGs 14 and 15 remain stagnant (see Figures 8 and 10). Additionally, only five to six HEIs have targeted SDG-6 from 2021 to 2024, reflecting the country's significant challenges and stagnant progress (see Figure 11). This again demands the utmost priority and a proactive role in addressing the SDGs of SPA-3.

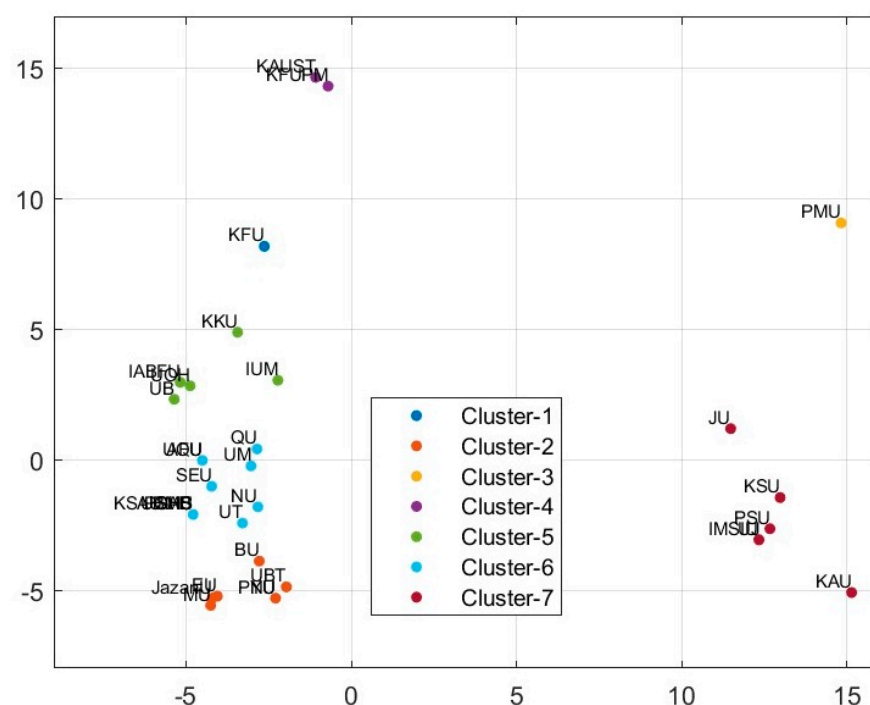


Figure 11. Two-dimensional hierarchical scatter plot of 33 HEIs of KSA using PCA.

6.2.4. SPA-4

The countrywide progress for these SDGs has already been covered in Section 6.1 and is not repeated here to avoid redundancy.

6.3. Key Findings and Conclusion for Hypothesis 3

The key findings presented in Sections 6.1 and 6.2 indicate a negative or weak correlation between the Saudi Arabian SD Report 2024 and the UN-SDGs prioritized by HEIs in THE IR. This noticeable gap in priority between HEIs and government entities could hinder collective SDG progress and demand HEIs to enhance their productive engagement and framework alignment. Moreover, a lack of comprehensive data and participation related to SDGs 1 and 10 presents a risk to national and institutional efforts to reduce poverty and inequality. Although the hypothesis anticipated a positive correlation between the performance of KSA HEIs in the EUA and UN-KSA Cooperation frameworks and the national SD priorities outlined in the 2024 SDR, the findings suggest a discrepancy, with

both exhibiting underperformance in addressing SDGs, thus providing partial support to study Hypothesis 3.

7. Cluster Identification of HEIs in Relation to UN-SDGs Coverage

In this analysis, hierarchical clustering is used to identify patterns among HEIs in Saudi Arabia based on their coverage of UN SDGs. To simplify data visualization and interpretation, principal component analysis (PCA) is also applied for dimensionality reduction, ensuring that the primary patterns within the data are preserved. The optimal number of clusters is determined using the silhouette method, which evaluates how closely related the data points are within each cluster while distinguishing between different clusters, thus enhancing insightfulness. Seven distinct clusters are identified and illustrated in Figure 11, with details provided in Table 6.

Table 6. Clustering distribution of 33 HEIs of KSA using HCA.

Cluster	No. of HEIs	Names of HEIs
1	1	KFU
2	7	PNU, JazanU, BU, MU, EU, UBT, YU
3	1	PMU
4	2	KFUPM, KAUST
5	5	IUM, IABFU, KKU, UOH, UB
6	11	UQU, QU, KSAU-HS, NU, UT, PSAU, SEU, UOHB, AFU, BMC, UM
7	6	IMSIU, KSU, KAU, JU, UJ, PSU

Additionally, Figure 12 presents a dendrogram depicting the hierarchical clustering of 33 Saudi HEIs, where the x -axis represents the HEIs, and the y -axis shows the dissimilarity or distance between clusters. Clusters are formed by merging HEIs with similar SDG coverage patterns, as indicated by the height at which two clusters converge.

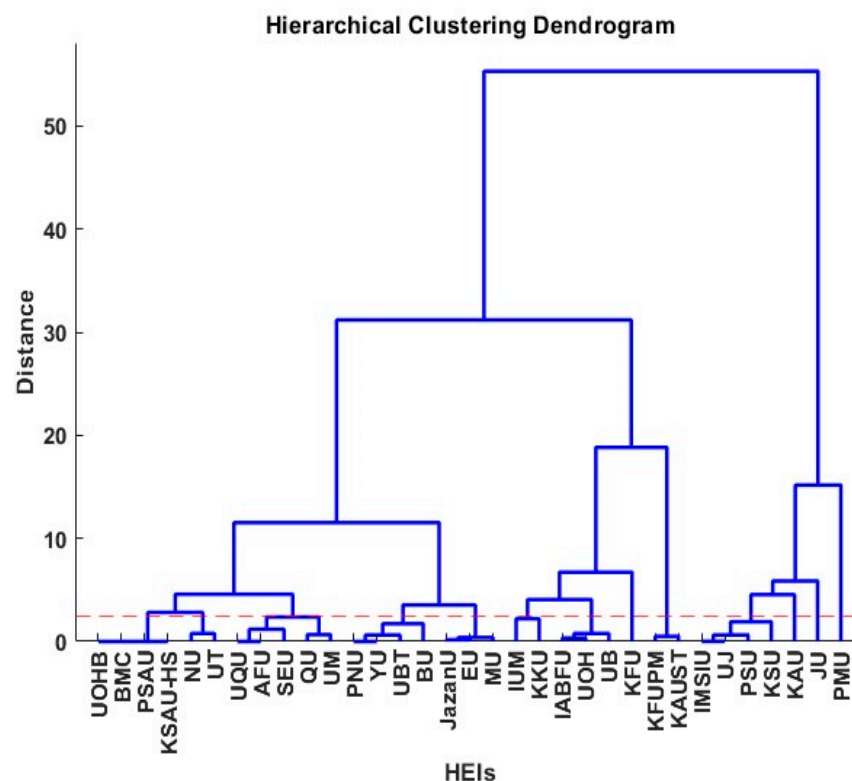


Figure 12. Hierarchical clustering dendrogram of 33 HEIs of KSA.

7.1. Key Insights and Strengths

The following subsections highlight unique group characteristics within the primary clusters shown in Figures 11 and 12, revealing the diversity in SDG coverage among Saudi HEIs.

7.1.1. Fully Aligned Groups

Four groups with fully aligned SDG focuses are identified from the dendrogram as follows:

Group-1: PNU and YU covering SDGs 4, 5, 8, and 10.

Group-2: KSAU-HS, PSAU, UOHB, and BMC, covering SDGs 3, 4, and 5.

Group-3: UQU and AFU covering SDGs 3, 4, and 9.

Group-4: IMSIU and UJ, covering SDGs 4, 5, 8, and 16.

7.1.2. Closely Aligned Groups

Four closely aligned groups are also identified, with each HEI group sharing focus areas:

Group-1: KFUPM and KAUST primarily focused on SDGs 3, 5, and 10.

Group-2: EU and JazanU covering SDGs 3, 5, and 10.

Group-3: NU and UT covering SDGs 3, 4, 5, and 8.

Group-4: QU and UM covering SDGs 3, 4, and 8.

Each of these HEIs covers additional SDGs beyond those listed.

7.1.3. Diversified Groups

PMU and KFU demonstrate broad SDG engagement, covering six and seven SDGs, respectively, since their participation in THE IR.

7.1.4. Focused Groups

Three focused groups are identified, one from Cluster-2 and two from Cluster-6:

- HEIs such as AFU and UQU excel in SDG 3, 4, and SDG 9.
- HEIs such as KSAU-HS, PSAU, UOHB, and BMC also show strong focus on SDGs 3, 4, and 5.

7.1.5. Consistently Performing HEIs

Some HEIs demonstrate consistent SDG performance, maintaining coverage over a minimum of four years. Notable examples include the following:

- AFU, consistently active for six years, covering SDGs 3, 4, and 9.
- UQU, consistently active for four years, covering SDGs 3, 4, and 9.
- PSU, consistently active for four years, covering SDGs 5, 8, and 16.
- UBT, consistently active for four years, covering SDGs 5, 8, and 10.

7.2. Proposed Grouping of HEIs for Enhanced and Advanced Coverage of UN-SDGs

For enhanced coverage and productive alignment of Saudi Arabian HEIs with the (EAU and UN-KSA Cooperation) frameworks, Table 7 lists leading HEIs recommended for each UN-SDG based on their frequency of participation, ranking performance (see Section 4), and identified clusters. Additionally, these leading HEIs are paired with one private and one public sector HEI to fully utilize potential and resources through Public–Private HEI Partnerships (PPHEIPs) [52]. In cases where no private HEI is found covering an SDG, only a public sector HEI is chosen as the leader. HEIs are also encouraged to take a proactive role in advancing UN-SDGs. Furthermore, potential group members for each SDG are listed along with the desired actions.

Table 7. Proposed grouping of 33 HEIs in KSA for each SDG, with designated lead HEIs.

SDG	Lead HEIs	Role	Potential Group Members	Desired Action
1	KFU, UM	Pioneering	UOH, KKU, IMSIU, QU, NU	A pioneering role is demanded from both KFU and UM nationwide, even though there are no data to measure progress (SDR-2024, KSA [5]).
2	KFU	Seek more collaborators, strengthen and lead	KKU	Opportunity for all HEIs, with a specific invitation to private sector HEIs, as they have not yet addressed this SDG.
3	AFU, IABFU	Strengthen and lead	UQU, PSAU, BMC, BU, UB, KSAU-HS, UOHB, JazanU	There is a clear opportunity for a strong Public–Private HEI partnership, as these SDGs are extensively targeted by both public and private sector HEIs. HEIs should leverage this strength to qualify for the top-tier ranking band B1. SDG-5 demands a pioneering role from PNU, taking the forefront, initiating actions, and making a significant impact.
4	QU, AFU	Strengthen and lead	UQU, UT, SEU, IUM, BMC, PSAU, KSAU-HS, NU, UB	
5	PNU, PSU	Pioneering	EU, UBT, UJ, JazanU, MU, KSAU-HS, PSAU, UOHB	
6	KAUST, PMU	Pioneering	KFU, KKU, KFUPM, QU	
7	KFU, KKU	Benchmarked	KFUPM, KAUST, IABFU, JU, UOH, UB	Opportunity for impactful partnerships. All four B1 HEIs are listed.
8	PNU, UM	Benchmarked	KAU, IUM, PSU, UJ, UBT, BU, PMU, YU	
9	KAU, AFU	Benchmarked	KSU, KFUPM, UQU, UOH	
10	KAU, UBT	Benchmarked	PNU, BU, MU, SEU, EU, YU, PMU, JazanU	A strong group must address the data availability gap under the leadership of KAU through proper planning and initiatives (SDR-2024, KSA [50]).
11	KAU	Strengthen and lead	KSU, KAUST	Become sustainable campuses. Target SCI [53]. Demand more HEIs on board. Opportunity for all HEIs.
12	KFU, PMU	Strengthen and lead	IUM, SEU	
13		not covered yet		Invitation and opportunity for all HEIs.
14	KFUPM, PMU	Seek more collaborators, strengthen and lead	KAUST	Joint Research Ventures with NEOM (Red Sea). Demand more HEIs on board.
15	UT	Seek more collaborators, strengthen and lead	None	Opportunity for all HEIs.
16	KAU, PSU	Strengthen and lead	IMISU, KSU, JU, UJ, PMU	There is a clear opportunity for a strong Public–Private HEI partnership.
17			Mandatory SDG covered by all HEIs	

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The relevant data for this study is given in Appendix A with a link for access and validation.

Acknowledgments: Author is grateful to all anonymous reviewers for constructive feedback which helped enhance the effectiveness of this research work.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

HEIs	Higher Educational Institutions
UN-SDGs	United Nations Sustainable Development Goals
THE IR	Times Higher Education Impact Rankings
EUA	European Union Association
UN-KSA	United Nations Sustainable Development Cooperation Framework with the Kingdom of Saudi Arabia 2022–2026
SDR	Sustainable Development Report
OECD	Organization for Economic Co-operation and Development
INES	Indicators of Education Systems
SPA	Strategic Priority Area
STAR	Sustainability Tracking, Assessment, and Rating System
HESI	Higher Education Sustainability Initiative
UIGWUR	UI GreenMetric World University Ranking
SET4HEI	Sustainability Evaluation Tool for Higher Education Institutions
EAUC	Environmental Association for Universities and Colleges
LIFE	Learning in Future Environments
AASHE	Association for the Advancement of Sustainability in Higher Education

Appendix A

This section contains Tables A1–A6 of Saudi Arabian HEIs participation in THE IR for years spanning 2019 until 2024 inherited from THE IR website link given below [21] <https://www.timeshighereducation.com/world-university-rankings> (accessed on 1 December 2024).

Table A1. Performance of KSA HEIs in THE Impact Rankings: 2019.

Achieved Bands	HEIs Name	UN SDGs Covered						
		3	4	5	9	10	11	17
54th	KAU			✓		✓	✓	✓
101–200	AFU	✓	✓		✓			✓
	KSU	✓		✓	✓			✓
Percentage Coverage by Participating HEIs		67	33	68	67	33	33	100

Table A2. Performance of KSA HEIs in THE Impact Rankings: 2020.

Achieved Bands	HEIs Name	UN SDGs Covered									
		1	2	3	4	5	8	9	10	16	17
101–200	KAU						✓	✓		✓	✓
401–600	AFU			✓	✓			✓			✓
	PNU					✓	✓		✓		✓
601+	KKU	✓	✓		✓						✓
	QU			✓	✓	✓					✓
Percentage Coverage by Participating HEIs		20	20	40	60	40	40	40	20	20	100

Table A3. Performance of KSA HEIs in THE Impact Rankings: 2021.

Achieved Bands	HEIs Name	UN SDGs Covered												
		1	3	4	5	6	7	8	9	10	12	14	16	17
27th	PNU				✓			✓		✓				✓
46th	KAU							✓	✓	✓				✓
101–200	PMU					✓		✓			✓			✓
201–300	UM	✓	✓					✓						✓
301–400	AFU		✓	✓					✓					✓
	KKU		✓				✓	✓						✓
	UQU		✓	✓					✓					✓
401–600	PSU				✓			✓					✓	✓
601–800	IABFU		✓	✓	✓									✓
	KFU			✓		✓	✓							✓
801–1000	KFUPM						✓		✓			✓		✓
	QU		✓	✓	✓									✓
Percentage Coverage by Participating HEIs		8	50	42	33	17	25	50	33	17	8	8	8	100

Table A4. Performance of KSA HEIs in THE Impact Rankings: 2022.

Achieved Bands	HEIs Name	UN SDGs Covered												
		1	3	4	5	6	7	8	9	10	12	14	16	17
4th	KAU			✓					✓	✓				✓
101–200	PMU					✓		✓			✓			✓
	PNU			✓	✓			✓						✓
201–300	IMSIU	✓		✓	✓									✓
	UM		✓	✓				✓						✓
	KAUST					✓	✓					✓		✓
	KFU						✓	✓			✓			✓
	PSU				✓			✓					✓	✓
301–400	KFUPM						✓		✓			✓		✓
	KKU					✓	✓	✓						✓
401–600	IABFU		✓		✓		✓							✓
	JU		✓	✓				✓						✓
	SEU		✓	✓						✓				✓
	UQU		✓	✓					✓					✓
601–800	AFU		✓	✓					✓					✓
	UOH	✓	✓						✓					✓
	PSAU		✓	✓	✓									✓
801–1000	BU		✓					✓		✓				✓
	QU			✓	✓	✓								✓
1000+	UBT				✓			✓		✓				✓
	UJ				✓			✓					✓	✓
	UT		✓	✓	✓									✓
Percentage Coverage by Participating HEIs		9	45	50	41	18	23	46	23	18	9	9	9	100

Table A5. Performance of KSA HEIs in THE Impact Rankings: 2023.

Achieved Bands	HEIs Name	UN SDGs															
		1	2	3	4	5	6	7	8	9	10	12	14	15	16	17	
101–200	KAUST						✓	✓					✓			✓	
	KFU	✓	✓					✓								✓	
	PMU								✓		✓		✓			✓	
201–300	UM	✓		✓					✓							✓	
	IABFU			✓		✓	✓									✓	
	IMSIU				✓				✓						✓	✓	
	KFUPM								✓	✓			✓			✓	
301–400	PSU					✓			✓						✓	✓	
401–600	KKU			✓			✓	✓								✓	
	QU	✓			✓				✓							✓	
601–800	AFU			✓	✓					✓						✓	
	BMC			✓	✓	✓										✓	
	PSAU			✓	✓	✓										✓	
	SEU				✓						✓	✓				✓	
	BU			✓	✓				✓							✓	
801–1000	UB			✓	✓			✓								✓	
	UOH	✓		✓	✓											✓	
	UJ				✓	✓			✓							✓	
	UQU			✓	✓					✓						✓	
	YU					✓			✓		✓					✓	
1001+	UBT					✓			✓		✓					✓	
	EU			✓		✓					✓					✓	
	IUM				✓				✓			✓				✓	
	NU			✓	✓	✓										✓	
	UT				✓				✓					✓		✓	
	Percentage Coverage by Participating HEIs		16	4	48	56	36	12	16	48	12	20	8	12	4	8	100

Table A6. Performance of KSA HEIs in THE Impact Rankings: 2024.

Achieved Bands	HEIs Name	UN SDGs															
		1	2	3	4	5	6	7	8	9	10	11	12	14	15	16	17
81	KFUPM						✓		✓	✓							✓
99	KFU	✓	✓					✓									✓
101–200	KAUST						✓					✓		✓			✓
	PMU						✓						✓			✓	✓
	IABFU			✓	✓		✓										✓
201–300	KAU								✓	✓						✓	✓
301–400	QU				✓	✓			✓								✓
	IMSIU				✓	✓			✓								✓
	UM			✓	✓				✓								✓
	PSU					✓			✓							✓	✓
401–600	KKU			✓			✓	✓									✓
	AFU			✓	✓					✓							✓

Table A6. Cont.

Achieved Bands	HEIs Name	UN SDGs																
		1	2	3	4	5	6	7	8	9	10	11	12	14	15	16	17	
601–800	UT				✓				✓						✓		✓	
	BMC			✓	✓	✓											✓	
	PSAU			✓	✓	✓											✓	
	JU			✓				✓								✓	✓	
	SEU			✓	✓						✓						✓	
801–1000	KSAU-HS			✓	✓	✓											✓	
	KSU			✓								✓				✓	✓	
	MU				✓	✓					✓						✓	
	UJ					✓			✓							✓	✓	
	UQU			✓	✓					✓							✓	
1001–1500	YU				✓				✓		✓						✓	
	JazanU			✓		✓					✓						✓	
	EU				✓	✓					✓						✓	
	UB			✓	✓			✓									✓	
	BU			✓					✓		✓						✓	
	UOH			✓	✓			✓									✓	
	UBT					✓			✓		✓						✓	
1501+	IUM				✓				✓				✓				✓	
	NU	✓			✓				✓								✓	
	UOHB			✓	✓	✓											✓	
Percentage Coverage by Participating HEIs		6	3	50	59	38	16	16	38	13	25	6	6	3	3	19	100	

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